Lilly Sweet

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EDUCATION

University of California, Berkeley MS/PhD in Mechanical Engineering — Aug 2025 – Dec 2026

GPA: 3.92 **Graduate Coursework:** Continuum, Rheology, Adv. CFD & Heat Transfer, Adv. Thermo, Adv. Fluids

University of Kansas BS in Mechanical Engineering — Jan 2021 – May 2025

GPA: 3.98 *(with distinction)* **Involvement:** Aero: Formula SAE

WORK EXPERIENCE

SpaceX Aero/Thermal Engineer Hawthorne, CA — Sep 2026 – Present

- Conducting aerodynamics/thermal analysis for the Falcon Rocket, Dragon Capsule, Starfall, and special projects.

SpaceX Falcon Graduate Engineer Hawthorne, CA — June 2026 – Aug 2026

- Executed **FUN3D CFD** simulations to obtain pressure and heat maps for a quick disconnect aero ramp and V4 Mini tile cover. Ran thermal analysis using CFD heat maps to confirm/size **TPS systems** for both hardware.
- Preprocessed, executed, and post-processed FUN3D CFD simulations simulating the wind tunnel experiments with multiple angles of attack. Found high corroboration between **wind tunnel** and CFD results. Tared out the CFD's sting impacts on aerodynamic coefficients, found agreement between Starfall CFD and flight data.
- Developed a Newtonian theory MATLAB script to analyze stability on simplified Starshield satellite models, enabling aero-stability analysis across multiple configurations to assess atmospheric demise.

SpaceX Raptor Graduate Engineer Hawthorne, CA — May 2025 – Aug 2025

- Integrated an **AI camera system** to automate machining process inspections, eliminating human-eye subjectivity and achieving **5–20× faster** throughput with improved repeatability and reliability. Utilized RobotStudio.
- **Automated** ingestion and organization of **nozzle extension test data** into an MQTT-backed database, enabling engineers to rapidly query results and validate propulsion hardware quality.

Tesla Controls Engineering Intern Fremont, CA — May 2024 – Aug 2024

- Led a **PLC migration** effort covering >$200K in hardware and labor, minimizing downtime while modernizing controls infrastructure for long-term reliability.

Burns & McDonnell Mechanical Engineering Intern Kansas City, MO — May 2023 – Aug 2023

- Developed **automated engineering workflow tools** by combining Python and PowerShell scripting, enabling unattended execution of complex analysis routines previously performed manually or in Excel.

US Army Combat Medic Sergeant (M-Day) — Jan 2020 – Jan 2026

- **Security Clearance:** Secret **Rank:** Sergeant (E-5) (Non-Commissioned Officer) **Honorable Discharge**

ENGINEERING PROJECTS

- **2D Viscous (DNS) with Unstructured Meshing FVM Solver:** A custom solver that simulates supersonic flow over a NACA0012. Roe's flux, Superbee & MC limiters (2^{nd} order w/ LS), able to successfully resolve shocks.
- **n^{th} Order FVM Euler Solver:** Used spectral, FR/CPR, and DG methods to achieve n^{th} order accuracy on 1D Euler flow through a converging/diverging nozzle.
- **FSAE Aerodynamics:** Ran iterative STAR-CCM+ simulations. Improved the team's process for higher fidelity simulations. Mechanically designed and manufactured mounting and wings.

SKILLS

- **Analysis:** STAR-CCM+, Patran/Nastran, Ansys (Mechanical, Fluent, Thermal), Thermal Desktop, Pointwise
- **Coding:** Python, MATLAB, C++/Arduino, Machine Learning, HPC
- **CAD/General:** SolidWorks, NX, ABB Robot Studio, PLC's (ABB, Siemens), Ignition, Carbon Fiber Layups